

Test Form A**Name** _____ **Date** _____**Chapter 11****Class** _____ **Section** _____1. Find a unit vector in the direction of $\mathbf{v}$ if $\mathbf{v}$ is the vector from $P(1, 2, -3)$ to $Q(4, -1, 0)$.

- (a) $3\mathbf{i} - 3\mathbf{j} + 3\mathbf{k}$ (b) $\mathbf{i} - \mathbf{j} + \mathbf{k}$ (c) $-\frac{1}{\sqrt{3}}\mathbf{i} + \frac{1}{\sqrt{3}}\mathbf{j} - \frac{1}{\sqrt{3}}\mathbf{k}$
- (d) $\frac{1}{\sqrt{3}}\mathbf{i} - \frac{1}{\sqrt{3}}\mathbf{j} + \frac{1}{\sqrt{3}}\mathbf{k}$ (e) None of these

2. Calculate $\mathbf{j} \times \mathbf{i}$.

- (a) $-\mathbf{j}$ (b) $-\mathbf{k}$ (c) $\mathbf{k}$ (d) $\mathbf{i}$ (e) None of these

In problems 3 through 8, let $\mathbf{u} = 3\mathbf{i} - \mathbf{j} - 2\mathbf{k}$, $\mathbf{v} = -2\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$ and $\mathbf{w} = \mathbf{i} + 2\mathbf{k}$.3. Calculate $\mathbf{u} \cdot \mathbf{v}$.

- (a) -7 (b) -13 (c) $\langle -8, -2, -11 \rangle$
- (d) 1 (e) None of these

4. Calculate $\mathbf{u} \times \mathbf{v}$.

- (a) $\langle 4, 10, -11 \rangle$ (b) $\langle -8, -2, -11 \rangle$ (c) -7
- (d) $\langle -4, -10, 11 \rangle$ (e) None of these

5. Calculate $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$.

- (a) -30 (b) $-18\mathbf{i} - 6\mathbf{j} - 6\mathbf{k}$ (c) -16
- (d) -24 (e) None of these

6. Calculate $\text{proj}_{\mathbf{v}}\mathbf{u}$.

- (a) $-\frac{7}{17}$ (b) $\langle \frac{14}{17}, \frac{21}{17}, -\frac{14}{17} \rangle$ (c) $\sqrt{17}$
- (d) $\langle 14, 21, -14 \rangle$ (e) None of these

7. Calculate $\cos \theta$ where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$.

- (a) $\frac{-7}{2\sqrt{17}}$ (b) $\frac{-7}{\sqrt{238}}$ (c) $\frac{-13}{\sqrt{238}}$ (d) $\frac{-13}{2\sqrt{17}}$ (e) None of these

8. Which of the following vectors is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$?

- (a) $\mathbf{k}$ (b) $-8\mathbf{i} + 2\mathbf{j} - 11\mathbf{k}$ (c) $\mathbf{v} \times \mathbf{u}$
- (d) $\mathbf{w}$ (e) None of these

9. Which of the following is an orthogonal pair of vectors?

- (a) $2\mathbf{i} - \mathbf{j}, \mathbf{i} + \mathbf{k}$ (b) $\mathbf{i} - \mathbf{j} + 2\mathbf{k}, -\mathbf{i} - \mathbf{j} - \mathbf{k}$ (c) $3\mathbf{i} - 2\mathbf{k}, 2\mathbf{j} - \mathbf{k}$
 (d) $5\mathbf{i} + \mathbf{j} + \mathbf{k}, -\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ (e) None of these

10. Which of the following is a set of parametric equations for the line through the points $(-2, 0, 3)$ and $(4, 3, 3)$?

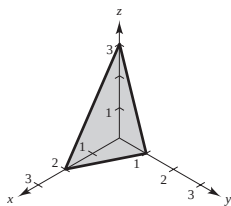
- (a) $x = -2 + 2t$
 $y = t$
 $z = 3 + t$ (b) $x = -2 + 2t$
 $y = 3t$
 $z = 3$ (c) $x = -2 + 2t$
 $y = t$
 $z = 3$
 (d) $x = -2 + 4t$
 $y = 3t$
 $z = 3 + 3t$ (e) None of these

11. Write an equation of the plane that contains the line given by $\frac{x}{1} = \frac{y-1}{3} = \frac{z+1}{2}$ and is perpendicular to the line given by $\frac{x-1}{-17} = \frac{y+5}{1} = \frac{z-3}{7}$.

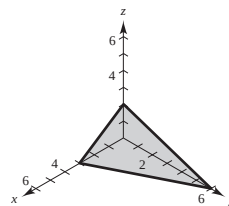
- (a) $17x - y - 7z - 6 = 0$ (b) $x + 3y + 2z + 8 = 0$ (c) $x + 3y + 2z - 1 = 0$
 (d) $17x - y - 7z + 1 = 0$ (e) None of these

12. Which of the following is a sketch of the plane given by $2x + y + 3z = 6$?

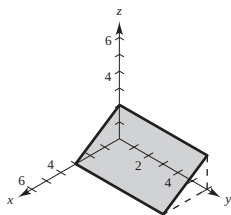
(a)



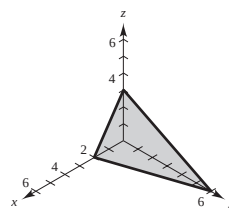
(b)



(c)



(d)



(e) None of these

13. Calculate the distance from the point $(-1, 2, 3)$ to the line given by $x = 2 - t$, $y = 1$, and $z = 1 + t$.

- (a) $\frac{5}{\sqrt{2}}$ (b) $\sqrt{\frac{3}{14}}$ (c) $\frac{5}{\sqrt{14}}$
 (d) $\sqrt{\frac{3}{2}}$ (e) None of these

14. Find the center and the radius of the sphere given by $x^2 + y^2 + z^2 + 2x - 2y + 6z + 7 = 0$.

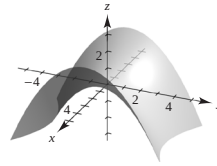
- (a) Center: $(1, 1, 3)$ Radius: 2
 (b) Center: $(-1, 1, -3)$ Radius: $\sqrt{37}$
 (c) Center: $(-1, 1, -3)$ Radius: 2
 (d) Center: $(1, -1, 3)$ Radius: $\sqrt{37}$
 (e) None of these

15. Identify the quadric surface given by $z = \frac{x^2}{4} + \frac{y^2}{16}$.

- (a) Elliptic cone
 (b) Elliptic paraboloid
 (c) Hyperbolic paraboloid
 (d) Hyperboloid of one sheet
 (e) Hyperboloid of two sheets

16. Identify the quadric surface sketched at the right.

- (a) Hyperboloid of two sheets
 (b) Elliptic paraboloid
 (c) Hyperboloid of one sheet
 (d) Elliptic cone
 (e) None of these



17. Find the equation of the surface of revolution if the generating curve, $y = 2x + 1$, is revolved about the y -axis.

- (a) $4x^2 - y^2 + 4z^2 + 2y - 1 = 0$
 (b) $3x^2 - z^2 + 4x + 1 = 0$
 (c) $4x^2 - y^2 - z^2 + 4x + 1 = 0$
 (d) $3x^2 - y^2 - z^2 + 4x + 1 = 0$
 (e) None of these

18. Express the cylindrical point $\left(4, \frac{\pi}{2}, 3\right)$ in spherical coordinates.

- (a) $\left(5, \frac{\pi}{2}, 60^\circ\right)$
 (b) $\left(5, \frac{\pi}{2}, \frac{3}{5}\right)$
 (c) $\left(5, \frac{\pi}{2}, 53\right)$
 (d) $\left(5, \frac{\pi}{2}, 0.93\right)$
 (e) None of these

19. Find an equation in spherical coordinates for the surface given by the rectangular equation $x^2 - y^2 = z$.

- (a) $\rho = \cot \phi \csc \phi$
 (b) $r^2 = z \sec 2\theta$
 (c) $\rho = \cot \phi \csc \phi \sec 2\theta$
 (d) $\rho = \tan \phi \csc \phi \sec 2\theta$
 (e) None of these

20. Find the work done in moving a particle from $P(3, -1, 0)$ to $Q(2, 3, 1)$ if the magnitude and direction of the force are given by $\mathbf{v} = \langle 5, 6, -2 \rangle$.

- (a) 5
 (b) 35
 (c) 17
 (d) 13
 (e) None of these